ENGINEERING PORTFOLIO

Jack Bajcz

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B.S. in Mechanical Engineering

M.S. in Mechanical Engineering

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ABOUT ME

Mechanical engineer with a strong foundation in large-scale cryogenic systems, I have excelled in designing and developing complex piping systems for research applications.

I have experience in every stage of the project development cycle, from ideation and process diagrams to detailed Solidworks assemblies and fabrication drawings. Each project has taught me valuable lessons about the design process, which I am excited to apply to future initiatives.

4K HELIUM DISTRIBUTION TRANSFER LINE

Description of Task:

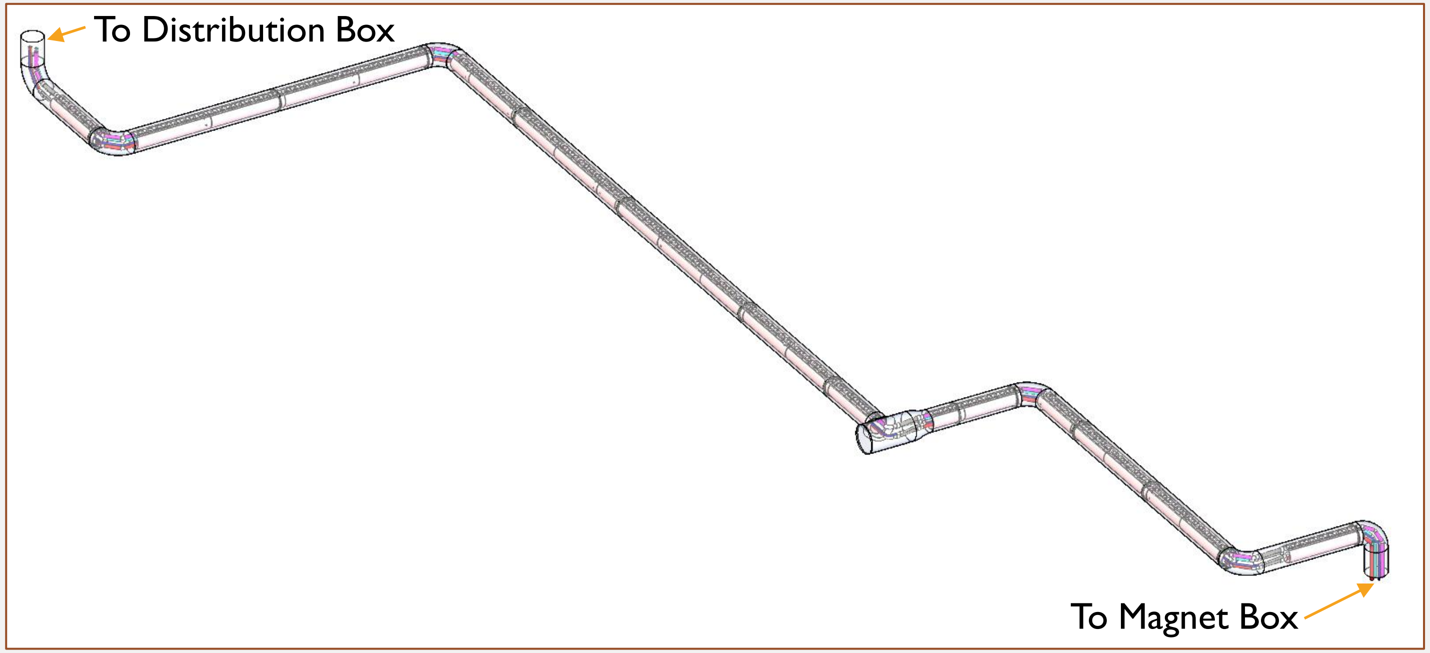
- Assisted in the design of a 15-meter long, multi-header cryogenic helium transfer line from the main distribution box to the magnets.

Scope of Work Completed:

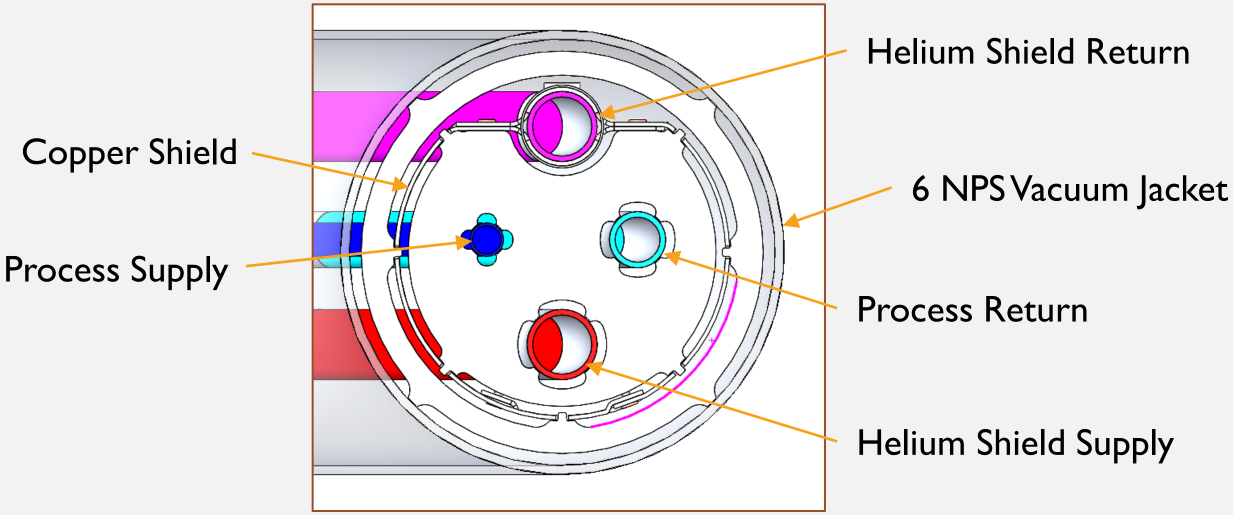
- Conducted preliminary flexibility calculations for 4K lines to ensure sufficient clearance between process lines.
- Modeled the transfer line assembly, including vacuum jacket, internal lines, rivets, shields, and spacers in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Drawings released for fabrication. System installation in progress.



Transfer Line Cross-Section



4K HELIUM MAGNET VALVE BOX AND WARM PIPING

Description of Task:

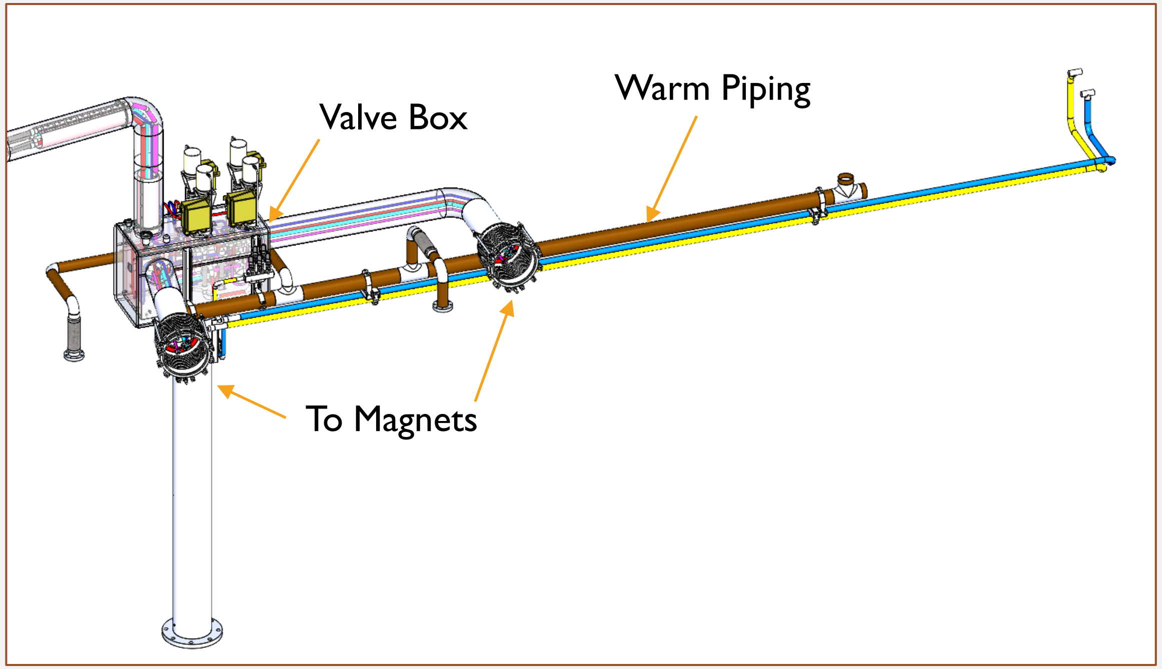
- Assisted in the design of a cryogenic helium magnet box and associated warm piping.

Scope of Work Completed:

- Developed a concept model for warm piping and supports.
- Created detailed models of valve box internals, vacuum jacket, magnet interfaces, and vacuum breaks in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Drawings released for fabrication. System installation in progress.



THESIS INSTRUMENTATION PIPING

Description of Task:

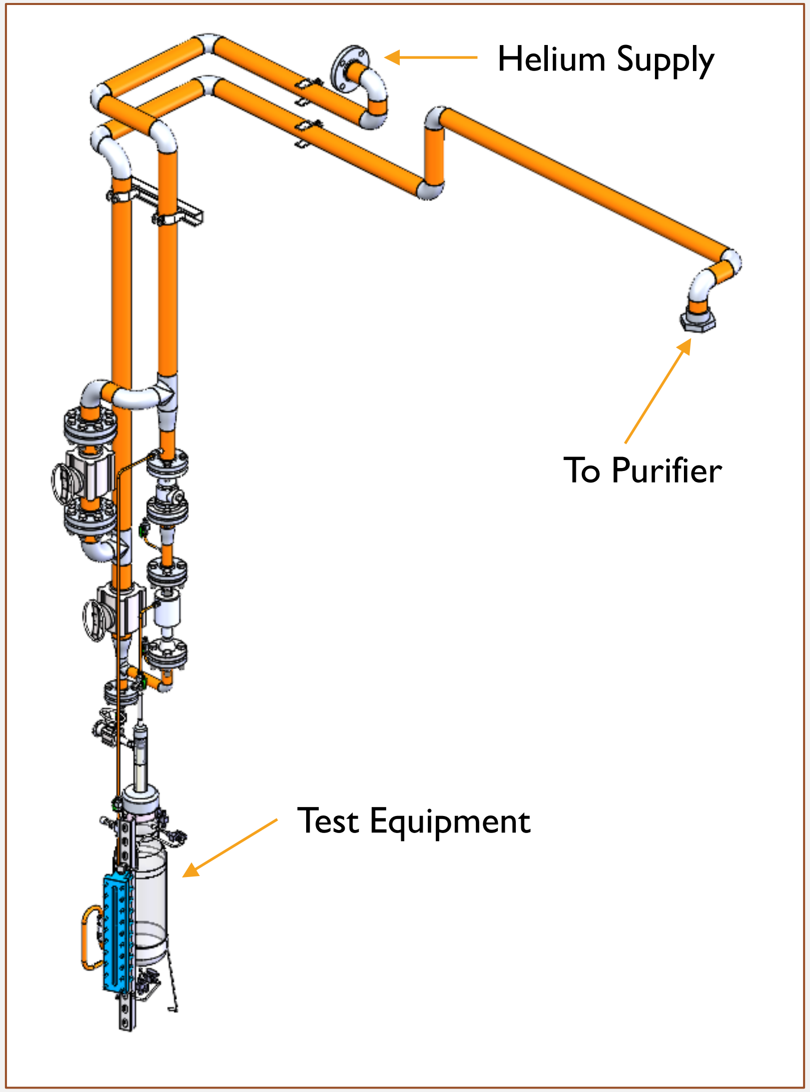
- Developed piping to support my master's thesis, testing a novel helium purifier design.
- Piping must be able to be installed and removed without changing existing connection points.

Scope of Work Completed:

- Designed pipe routing, instrumentation, and supports.
- Created detailed CAD model of full assembly in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Fabricated, installed, and in use for Helium Purifier Experiments



4K HELIUM DISTRIBUTION TRANSFER LINE AND MAGNET CONNECTIONS

Description of Task:

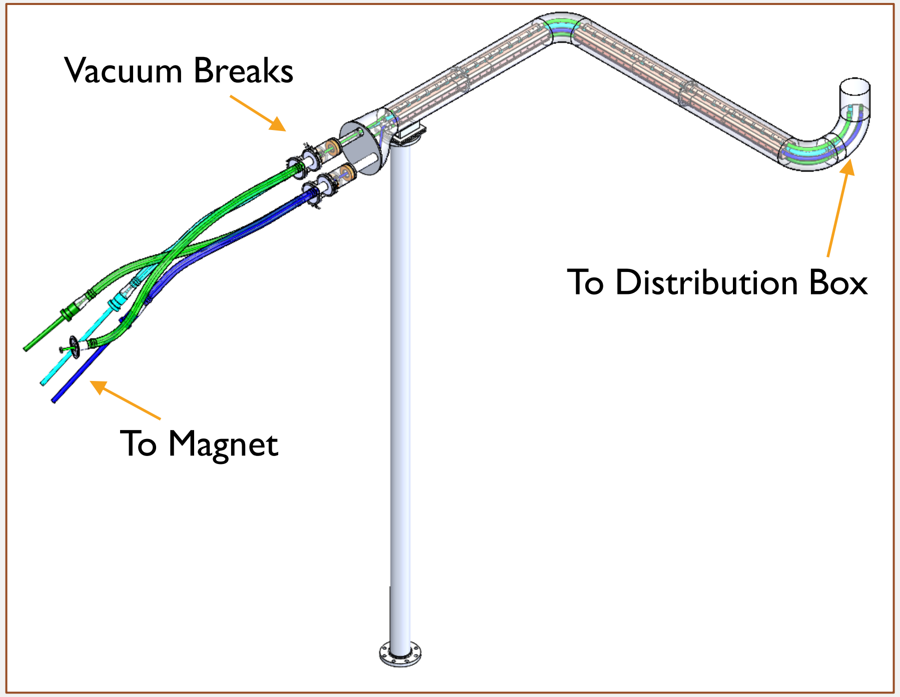
- Assisted in the design of a multi-header cryogenic helium transfer line with flex hose magnet connections.

Scope of Work Completed:

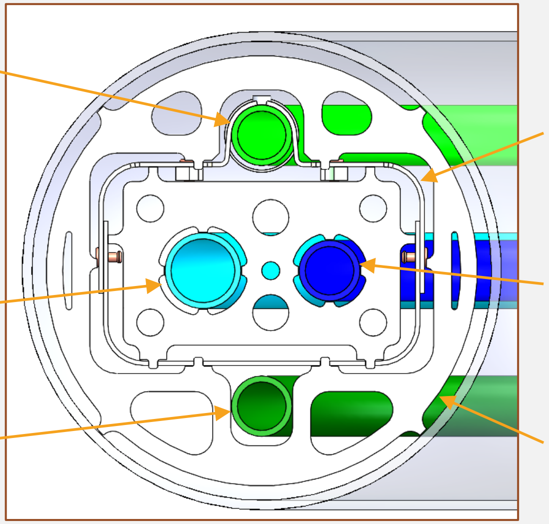
- Modeled the transfer line assembly, including vacuum jacket, internal lines, rivets, shields, and spacers in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Drawings released for fabrication.



Transfer Line Cross-Section



4K HELIUM DISTRIBUTION TRANSFER LINE AND MAGNET CONNECTIONS

Description of Task:

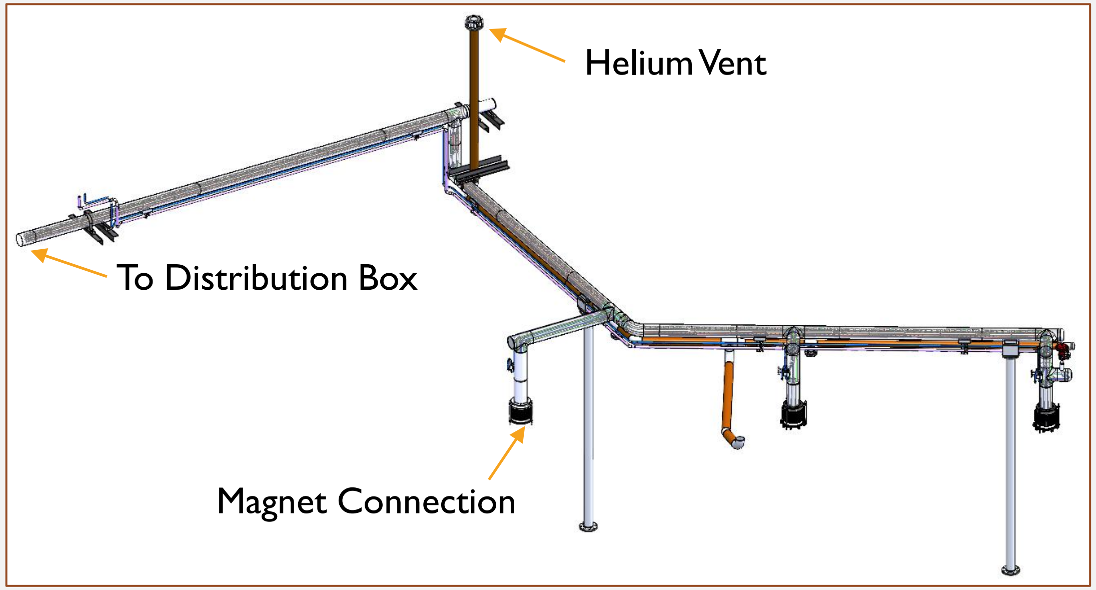
- Assisted in the design of a multi-header cryogenic helium transfer line and associated warm piping.

Scope of Work Completed:

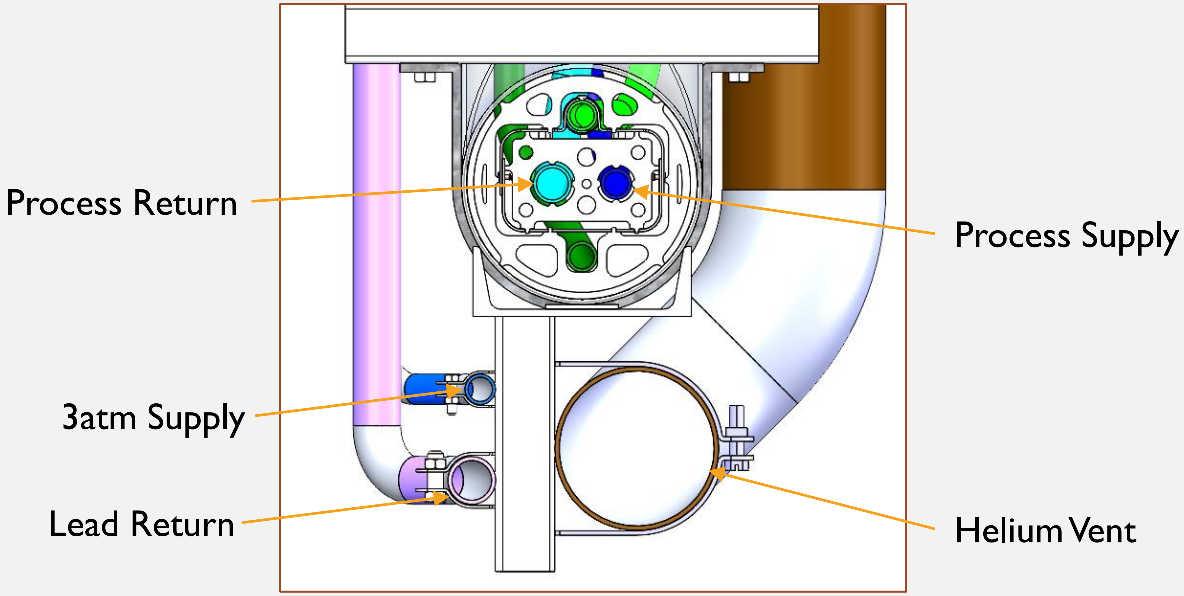
- Developed a concept model for warm piping and supports.
- Modeled the transfer line assembly, including vacuum jacket, internal lines, rivets, shields, and spacers in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Drawings released for fabrication.



Transfer Line Cross-Section



WARM GAS DISTRIBUTION LINES

Description of Task:

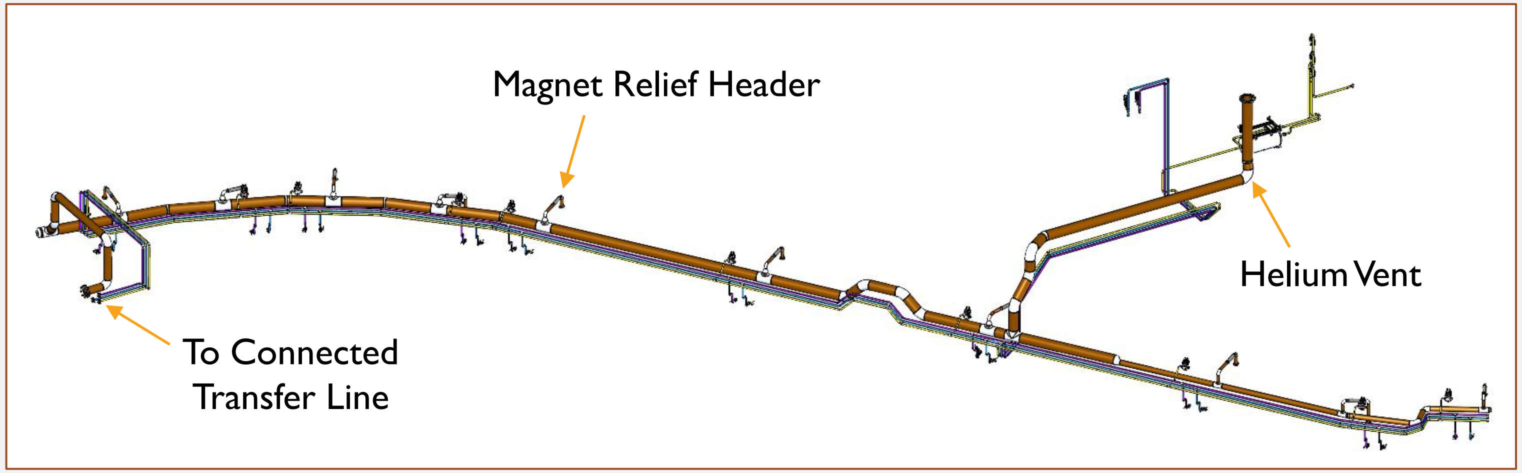
- Assisted in the design of a warm piping system to supply experiment instrumentation.
- System includes 4 separate 32-meter-long lines.

Scope of Work Completed:

- Developed a concept model for warm piping and supports.
- Created a detailed model in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Drawings released for fabrication. System installation in progress.



WARM GAS DISTRIBUTION LINES

Description of Task:

- Assisted in the design of a warm piping system.

Scope of Work Completed:

- Developed a concept model for warm piping and supports.
- Created a detailed model in Solidworks.
- Created fabrication drawings for all parts and assemblies.

Status of Project:

- Drawings released for fabrication.